

Product Recommendation via Rationale-Aware Gating over Sparse Review-Aspect Knowledge Graphs

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Abstract

RA-GARK is a knowledge-graph-aware recommender designed for sparse review-derived knowledge graphs. The method keeps collaborative filtering and KG semantics in two separate views: a LightGCN local view models the user–item graph, while a global view represents each item by SVD-initialized latent aspect slots and selects user-relevant slots with softmax rationale masking. A local-biased fusion gate then decides how much KG information enters the final score. The gate is initialized so that the model starts close to pure LightGCN and opens the KG channel only when training signal supports it. On a filtered Amazon Books benchmark, RA-GARK improves over both the strongest KG-aware baseline and the pure LightGCN anchor, showing that sparse KG signal can become useful when it is optional rather than mandatory.

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1 Introduction

1.1 Background and Motivation

Graph neural networks have become a central modeling tool for collaborative filtering. Among them, LightGCN [1] shows that a recommender can perform strongly with only normalized neighborhood propagation and layer-wise averaging, without nonlinear transformations or feature matrices. This simplicity makes LightGCN a strong collaborative baseline and a useful reference point for evaluating whether side information is truly beneficial.

Knowledge-graph-aware recommendation extends this setting by adding item-side semantics. Methods such as KGAT [2], KGCL [3], MCCLK [4], and KGRec [5] integrate KG structure through entity propagation, contrastive alignment, or rationale selection. These designs are effective when the KG is dense and reliable. However, real recommendation settings often rely on review-derived or domain-specific KGs whose coverage is sparse and uneven.

1.2 Sparse-KG Observation

This work begins from a simple empirical observation. On the Amazon Books review-derived KG used in this study, each item has only 2.4 aspect edges on average after filtering. Under the same split and evaluation protocol, all four KG-aware baselines fall below pure LightGCN on NDCG@20, as shown in Table 1. The result does not imply that these baselines are weak in general; rather, it indicates that mandatory KG integration can become harmful when the KG is too sparse to provide stable semantic signal.

Table 1: KG-aware baselines and pure LightGCN on the sparse review-derived KG.

Method	NDCG@20
MCCLK [4]	0.1067
KGCL [3]	0.1073
KGAT [2]	0.1079
KGRec [5]	0.1095
Pure LightGCN [1]	0.1179

1.3 Research Question and Contributions

The central question is how to use KG signal when it is helpful without letting KG noise contaminate the collaborative representation when it is not. RA-GARK addresses this by treating the KG as a gateable side channel rather than a mandatory component of the scoring path.

The contributions are threefold. First, we report a sparse-KG failure mode in which representative KG-aware methods underperform a no-KG LightGCN anchor. Second, we propose RA-GARK, a dual-view architecture with KG-SVD initialization, softmax rationale masking, and a local-biased fusion gate. Third, we show through main results, ablations, and a qualitative case study that KG semantics can improve ranking and reveal item characteristics when the architecture can regulate their influence.

2 Related Work

2.1 Graph Collaborative Filtering

LightGCN [1] simplifies graph collaborative filtering by removing feature transformation and nonlinear activation from GCN-style recommendation. It propagates embeddings over the user–item bipartite graph and averages layer outputs, making it efficient and stable. Later methods such as SGL [6] add self-supervised objectives on top of this collaborative backbone.

RA-GARK adopts LightGCN directly as its local view. This choice is deliberate: in the sparse KG setting studied here, the collaborative signal is the reliable default, so KG information should improve this baseline rather than be mixed into it by force.

2.2 KG-Aware Recommendation

KGAT [2] represents the deep-fusion approach: user–item interactions and KG triples are merged into a collaborative knowledge graph, and entity messages participate in user and item representation learning. KGCL [3] and MCCLK [4] add contrastive learning between KG and collaborative views, while KGRec [5] introduces edge-level rationale learning to identify important KG triples.

These methods assume that the KG is reliable enough to be used inside the main representation path. RA-GARK starts from a weaker assumption. It allows KG evidence to affect the score, but only through a side channel whose contribution is controlled by a learned gate.

2.3 Rationale and Gating

KGRec is the closest related method because it makes rationale selection explicit. The difference is the site and granularity of the rationale. KGRec selects KG edges inside a KGAT-style propagation path. RA-GARK selects among latent item-aspect slots in a separate global view and then decides whether the selected KG signal should enter the final score.

The fusion gate follows the same broad principle as Highway Networks [7]: initialize toward

a known safe default and let learning open the transformed path only when it helps. In RA-GARK, the safe default is pure LightGCN. The gate bias is initialized to $+5$, making the initial local-view weight approximately 0.993.

2.4 Attention Normalization

Attention normalization is not a cosmetic choice. DIN [8] uses sigmoid-style attention when historical behaviors can contribute independently, whereas NAIS [9] and AFM [10] use softmax when candidates compete. RA-GARK uses softmax over item-aspect slots because the rationale module selects which semantic facet of the item is most relevant to the user. The ablation results in Chapter 4 show that replacing this softmax head with sigmoid is the largest single-component degradation in our setting.

3 Proposed Method

3.1 Overview

RA-GARK consists of a local collaborative view, a global KG view, a local-biased fusion gate, and a BPR training objective with small cross-view contrastive regularization. Figure 1 summarizes the data flow. The local view is a standard LightGCN branch over the user-item graph. The global view represents each item by $A = 4$ latent aspect slots initialized from the review-derived KG. The two views are fused only at the final representation stage.

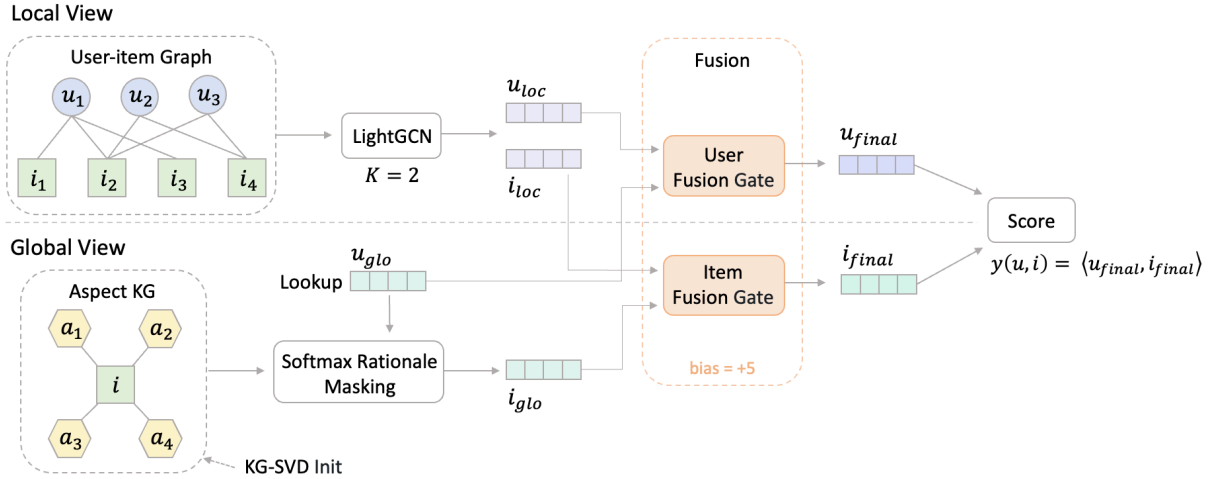


Figure 1: RA-GARK architecture.

3.2 Local View

Let $\mathbf{R}$ be the user-item interaction matrix and

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{0} \end{bmatrix}.$$

The normalized adjacency is $\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. Starting from trainable user and item embeddings $\mathbf{E}^{(0)}$, LightGCN propagates

$$\mathbf{E}^{(\ell+1)} = \tilde{\mathbf{A}}\mathbf{E}^{(\ell)}, \quad \ell = 0, \dots, K-1,$$

and averages the layers:

$$\bar{\mathbf{E}} = \frac{1}{K+1} \sum_{\ell=0}^K \mathbf{E}^{(\ell)}.$$

The corresponding rows give u_{loc} and i_{loc} . No KG signal enters this branch.

3.3 Global View

The global item representation is built from the item–aspect KG. We first form a binary item–aspect matrix $\mathbf{M}$ and apply IDF weighting to reduce the dominance of generic aspects. A rank- Ad truncated SVD is then computed, and the resulting item embeddings are reshaped into $\mathbf{a}_i \in \mathbb{R}^{A \times d}$, where $A = 4$ is the number of latent aspect slots and $d = 128$ is the embedding dimension.

For a user–item pair, the global user embedding u_{glo} is a direct lookup. The item-side rationale module scores each slot conditioned on the user and normalizes the scores with softmax:

$$w_{u,i,k} = \frac{\exp(s_{u,i,k}/\tau)}{\sum_{k'=1}^A \exp(s_{u,i,k'}/\tau)}.$$

The global item embedding is the weighted aspect mixture:

$$i_{\text{glo}} = \sum_{k=1}^A w_{u,i,k} \mathbf{a}_{i,k}.$$

The rationale therefore answers which latent item aspect is most relevant to the user for this recommendation.

3.4 Local-Biased Fusion Gate

RA-GARK uses separate user-side and item-side gates:

$$u_{\text{final}} = \alpha_u u_{\text{loc}} + (1 - \alpha_u) u_{\text{glo}}, \quad i_{\text{final}} = \alpha_i i_{\text{loc}} + (1 - \alpha_i) i_{\text{glo}}.$$

Both α_u and α_i are produced by small MLPs with sigmoid outputs. Their final-layer bias is initialized to +5, so $\sigma(5) \approx 0.993$. At the beginning of training, the model is therefore almost pure LightGCN; the KG channel opens only if gradients indicate that it improves ranking.

3.5 Training Objective

The prediction score is the inner product

$$\hat{y}(u, i) = \langle u_{\text{final}}, i_{\text{final}} \rangle.$$

Training uses Bayesian Personalized Ranking [11]:

$$\mathcal{L}_{\text{BPR}} = -\log \sigma(\hat{y}(u, i^+) - \hat{y}(u, i^-)).$$

Two small InfoNCE-style regularizers align the local and global views for users and items. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \lambda_{\text{CL}}(\mathcal{L}_{\text{aCL}} + \mathcal{L}_{\text{uCL}}),$$

with $\lambda_{\text{CL}} = 0.005$, so contrastive learning supports the geometry without becoming the main fusion mechanism.

4 Experiments and Results

4.1 Dataset and Setup

The experiments use a filtered Amazon Books subset with 905 users, 1,399 items, and 22,265 positive interactions. The review-derived item–aspect KG contains 3,370 edges over 2,098 unique aspects after filtering, giving only 2.4 aspect edges per item on average. Interactions are split by user into 70/15/15 train, validation, and test sets with seed 42.

Table 2: Dataset statistics after preprocessing.

Statistic	Value
#Users	905
#Items	1,399
#Interactions	22,265
#KG edges	3,370
#Unique aspects	2,098
Avg. edges per item	2.4

We compare RA-GARK with pure LightGCN and four KG-aware baselines: KGAT, KGCL, MCCLK, and KGRec. Evaluation uses full ranking, not sampled negatives. NDCG@20 is the primary metric, with HR, Recall, and MAP as secondary metrics.

4.2 Main Results

Table 3 reports the Top-20 result. RA-GARK achieves the best score on all four metrics. The most important comparison is against pure LightGCN: the four KG-aware baselines are below the no-KG anchor, while RA-GARK moves above it. This supports the claim that the same sparse KG can become useful when the model can control whether and how KG signal enters the score.

Table 3: Top-20 recommendation performance.

Model	NDCG@20	HR@20	Recall@20	MAP@20
MCCLK [4]	0.1067	0.4530	0.1720	0.0497
KGCL [3]	0.1073	0.4696	0.1827	0.0479
KGAT [2]	0.1079	0.4773	0.1807	0.0491
KGRec [5]	0.1095	0.4729	0.1834	0.0500
LightGCN [1]	0.1179	0.4917	0.1937	0.0555
RA-GARK	0.1243	0.4972	0.2020	0.0594
Δ vs KGRec	+13.5%	+5.1%	+10.1%	+18.8%
Δ vs LightGCN	+5.4%	+1.1%	+4.3%	+7.0%

4.3 Ablation Study

The ablation study removes one design component at a time. The largest drop comes from replacing softmax rationale masking, showing that attention normalization is a load-bearing design choice in this architecture. Removing KG-SVD initialization and removing the local-biased gate also reduce performance, which confirms that the KG channel must be both well initialized and carefully admitted.

Table 4: Ablation results under Top-20 evaluation.

Model	NDCG@20	MAP@20
RA-GARK (full)	0.1243	0.0594
w/o softmax head	0.1005	0.0451
w/o KG-SVD init	0.1171	0.0545
w/o fusion-gate bias	0.1194	0.0555
w/o MLP gate	0.1180	0.0552
w/o user CL	0.1192	0.0563
w/o aspect CL	0.1200	0.0570

4.4 Case Study and Cost

The qualitative case study shows why KG is not merely an auxiliary feature in RA-GARK. Because the global view selects item aspects conditioned on the user, the model can expose which semantic facets contribute to a recommendation. This supports the contribution that KG-based recommendation can reveal product characteristics beyond collaborative co-occurrence.

The additional computation is small. On the reported hardware, RA-GARK trains at approximately 1.5 seconds per epoch, comparable to LightGCN and KGRec at the same embedding dimension. The dominant cost remains LightGCN propagation; the aspect slots, rationale masking, and fusion gates add only small batch-level operations.

5 Conclusion and Future Work

5.1 Conclusion

This paper presented RA-GARK, a dual-view KG-aware recommender for sparse review-derived knowledge graphs. The main design principle is that KG should be a controllable side channel rather than a mandatory part of the scoring pipeline. The local view preserves a strong LightGCN collaborative representation, the global view encodes item semantics through SVD-initialized aspect slots and softmax rationale masking, and the fusion gate decides how much KG information enters the final user and item embeddings.

The experimental result shows that RA-GARK improves over both KGRec and pure LightGCN on the sparse Amazon Books benchmark. This is the key finding: the same KG that hurts several mandatory-fusion baselines becomes beneficial when its contribution is regulated by the architecture. The case study further suggests that the KG view helps the model distinguish items by semantic characteristics, not only by collaborative co-occurrence.

5.2 Limitations

The main limitation is the sparse KG regime itself. RA-GARK is calibrated for a setting where KG signal is unreliable and should be treated cautiously. On dense and clean KGs, the local-biased gate may be too conservative, and deep-fusion methods may be more appropriate. The quantitative evidence is also based on a single Amazon Books benchmark, so the attention-normalization finding and the gate-bias design should be validated on additional datasets.

5.3 Future Work

Future work should test RA-GARK on both sparse and dense KG benchmarks to identify when a KG side channel is preferable to deep fusion. Another direction is to adapt the gate bias to the data regime: when collaborative interactions are sparse but KG is dense, the safe default may need to shift toward the KG view instead of LightGCN. Finally, the distinction be-

tween softmax and sigmoid rationale heads should be replicated in other rationale-aware recommenders, because the present result suggests that attention normalization can be an architectural decision rather than a minor hyperparameter.

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